AI WeatherWise API

Intelligent Weather Forecasts and Insights

Project Report

1. Abstract

AI WeatherWise is a RESTful backend platform built with Node.js and Express.js, using MongoDB with the Mongoose ODM for data persistence. It delivers real-time weather metrics and enriches them with natural-language summaries and practical advice generated by Google Gemini. The platform exposes a secured, versioned API that client applications can use to display current conditions, forecasts and AI-driven insights.

2. Introduction

Raw weather data is often difficult for everyday users to interpret. AI WeatherWise bridges this gap by combining live meteorological data with a generative AI layer that explains what the numbers mean, for example what to wear, whether to carry an umbrella, or when outdoor plans are best scheduled.

3. Objectives

- Provide real-time weather data through a clean REST interface.
- Generate human-readable AI insights for each location.
- Persist users and search history in MongoDB.
- Secure endpoints with token-based authentication.
- Keep the API simple to integrate and easy to demonstrate.

4. Technology Stack

Layer	Technology
Runtime and framework	Node.js, Express.js
Database	MongoDB with Mongoose ODM
AI service	Google Gemini
Weather data	Real-time weather provider API
Security	Token (JWT) authentication

5. System Architecture

A client sends a request to the Express.js server. Middleware validates the token and parameters. The weather controller fetches live data from the weather provider, then passes the metrics to Google Gemini to generate an insight. The combined result is returned to the client, and the search is saved in MongoDB for history.

```
Client -> Express API -> Auth middleware -> Weather controller
                                          |-> Weather provider (live data)
                                          |-> Google Gemini (AI insight)
                                          '-> MongoDB (users, history)
```

6. API Reference

Base URL: https://api.your-domain.com/api/v1 Auth header: Authorization: Bearer <TOKEN>

Method	Endpoint	Description
POST	/auth/register	Create an account
POST	/auth/login	Obtain an access token
GET	/weather/current?city=	Real-time weather for a city
GET	/weather/forecast?city=&days=	Multi-day forecast
GET	/weather/insights?city=	AI summary and advice
GET	/history	Saved searches

Example request

```
curl -H "Authorization: Bearer <TOKEN>" \
  "https://api.your-domain.com/api/v1/weather/insights?city=Coimbatore"
```

Example response

```
{
  "city": "Coimbatore",
  "temperature": 29.4,
  "humidity": 68,
  "condition": "Partly cloudy",
  "insight": "Warm and humid. Carry water; light rain possible by evening."
}
```

7. Error Codes

Code	Meaning
400	Missing or invalid parameters
401	Missing or invalid token
404	City not found
429	Rate limit exceeded
500	Server or AI service error

8. Setup and Deployment

```
git clone <repo-url> && cd weatherwise
npm install
# .env: PORT, MONGO_URI, WEATHER_API_KEY, GEMINI_API_KEY, JWT_SECRET
npm start
```

9. Demo and Downloads

Live demo: https://drive.google.com/file/d/1-fkwwNVUkc5Q2v-vmiELzLbgHjr2PjKr/view?usp=drive_link

Code Drive:https://drive.google.com/file/d/1HVmSLbO3JIoG7lBcWtEKoSaV7-_axto4/view?usp=drive_link